SHONGJOG

সংযোগ | INCLUSIVE EMERGENCY RESPONSE & COORDINATION

A warning is only useful when everyone can act.

A bilingual, accessibility-aware coordination platform connecting citizens, emergency coordinators, verified resources and field responders.



Dacope coastal delta context — project visual asset.

CITIZEN INTAKE	COORDINATION	FIELD RESPONSE
Accessible assistance requests in Bangla or English, with functional needs, location and consent.	Case review, resource matching, capacity checks and human-authorized dispatch.	Mobile responder workflows, navigation hand-off and status updates.

PROJECT OVERVIEW • V4.2.1 • 07 NOVEMBER 2026

Live demo: https://shongjog-eight.vercel.app

Source: github.com/Safin313-REBORN/SHONGJOG-PLATFORM-OPERATIONAL-DIRECTIVE-FOR-ANTIGRAVITY

01 / EXECUTIVE OVERVIEW

One platform. Three operational roles. One inclusive response chain.

SHONGJOG is designed as a last-mile coordination layer for emergencies in coastal Bangladesh. It turns an unstructured assistance request into a reviewed, constraint-checked and traceable response workflow.

THE PROBLEM

A warning may reach a community without giving every person a practical way to evacuate. Access, transport, communication and support requirements can be overlooked.

THE RESPONSE

Capture functional needs, compare them with verified resource capabilities, reserve capacity transactionally and keep a human coordinator accountable for dispatch.

THE PRIMARY CONTEXT

A demonstration scenario focused on Dacope Upazila, Khulna, with coastal roads, waterways, shelters and volunteer resources represented in the product.

THE EXPERIENCE

One bilingual product for citizens, coordinators and field responders, with responsive layouts for phones, tablets and desktop command stations.

At a glance

Dimension	Project profile
Product category	Humanitarian emergency coordination / accessibility technology
Primary users	Citizens and caregivers; coordinators; responders; resource administrators
Geographic focus	Dacope, Khulna, Bangladesh (demonstration context)
Languages	English and Bangla, with Banglish input recognition described in the AI layer
Core stack	Next.js 16.4.5, React 19.3.0, TypeScript, Prisma 6.4.1, PostgreSQL
Deployment profile	Vercel application hosting with Neon PostgreSQL, per the current GitHub main branch
Product boundary	Demonstration and coordination prototype; not an official emergency dispatch authority

Important operating boundary

The interface labels the scenario as a DEMO SIMULATION and not a live emergency. The project should not be represented as a replacement for Bangladesh's official emergency number, disaster authorities or field-tested response organizations.

02 / WHY IT EXISTS

The last mile is a coordination problem.

Emergency access depends on more than sending an alert. People may need transport, communication support, a step-free route, caregiver accompaniment or continuity for powered assistive equipment.



Community consultation context



Cyclone-shelter infrastructure

Evidence that informs the design

84%	6%	6,342 RESPONSES
In the UNDRR 2023 global survey, 84% of respondents with disabilities reported having no personal disaster preparedness plan. [1]	Six per cent reported that they could not evacuate independently even when sufficient early warning was provided. [1]	The UNDRR survey collected 6,342 responses from 132 countries; these are global findings, not Bangladesh-only estimates. [1]

A 2026 study on disability-inclusive and gender-responsive early warning systems in Khulna highlights continuing barriers in accessibility, coordination, data and infrastructure. [2] SHONGJOG translates this problem framing into a practical workflow that records functional needs and checks resource suitability before dispatch.

Design principles derived from the problem

- Ask about functional barriers and assistance requirements, not unnecessary clinical diagnoses.
- Treat accessibility and verification as eligibility conditions, not optional ranking bonuses.
- Do not allow geographic proximity to override a physical safety requirement.
- Keep consequential dispatch decisions under accountable human review.
- Be transparent about simulation, missing information, offline drafts and unverified assets.

> **Interpretation note**
>
> The statistics above describe international survey respondents. They establish the importance of inclusive disaster planning; they should not be presented as measured prevalence rates for Dacope or all of Bangladesh.

03 / PRODUCT MODEL

How SHONGJOG connects the response chain

The product is organized around three working roles and a shared resource registry. Each role sees the information and actions needed for its part of the response.

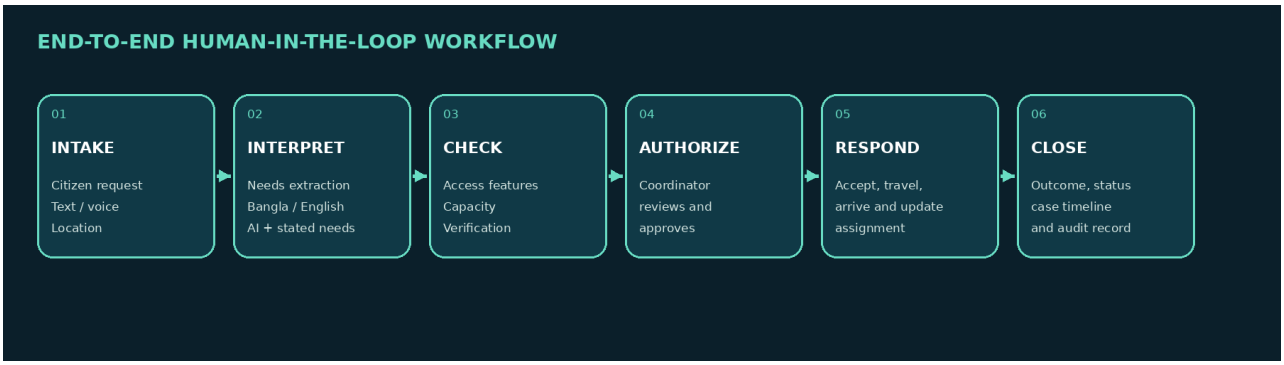


Figure 1. Intended request-to-closure workflow represented in the application.

Role responsibilities

CITIZEN / CAREGIVER Creates an assistance request, describes the situation, selects functional needs, shares a location and consent, then keeps the case reference.	**COORDINATOR** Reviews intake, checks extracted needs, examines matching explanations, chooses eligible resources and authorizes dispatch.
FIELD RESPONDER Receives a minimized operational briefing, accepts or declines an assignment, progresses through response states and records completion.	**RESOURCE ADMINISTRATOR** Maintains resource records, capacity, capabilities, locations and verification status so the matcher can work from structured data.

Core interaction loop

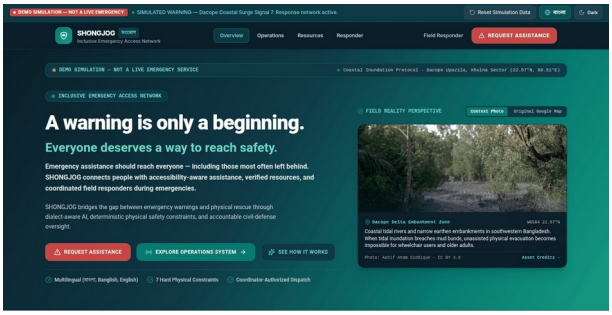
Request → needs → constraints → coordinator approval → assignment → responder updates → case timeline. The design aims to keep the same case context visible as responsibility passes between roles.

> **Why this matters**
>
> The platform is not just an intake form. Its value proposition is the connected operational lifecycle: a citizen request can be reviewed, matched, assigned, acted on and recorded in one system.

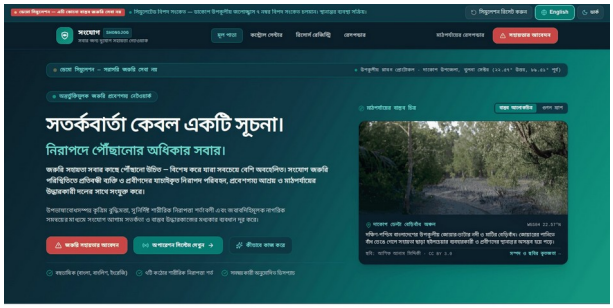
04 / CITIZEN EXPERIENCE

A request flow designed for stressful moments

The citizen experience is mobile-first and structured to reduce the effort needed to explain an emergency.



Landing page — English



Landing page — Bangla

Request intake sequence

Step	What the citizen provides	System response
1. Situation	Free-text description, spoken or typed where supported; approximate location and group size.	AI-assisted interpretation proposes language, urgency, support needs and missing information.
2. Functional needs	Checkbox or preset needs such as wheelchair transport, step-free access, guided assistance, sign language and power equipment.	Stated needs are combined with extracted needs for coordinator review.
3. Review & consent	Review the summary and confirm data-sharing consent before submission.	The API validates the description and consent, then creates a case and timeline events.
4. Tracking	Case reference and current status.	The case can be revisited by reference through the application's tracking experience.

Mobile request interface

The interface supports English and Bangla display. The AI layer is designed to recognize Bengali Unicode, Banglish transliteration and English. Browser voice input depends on browser/device speech-recognition support and permissions.

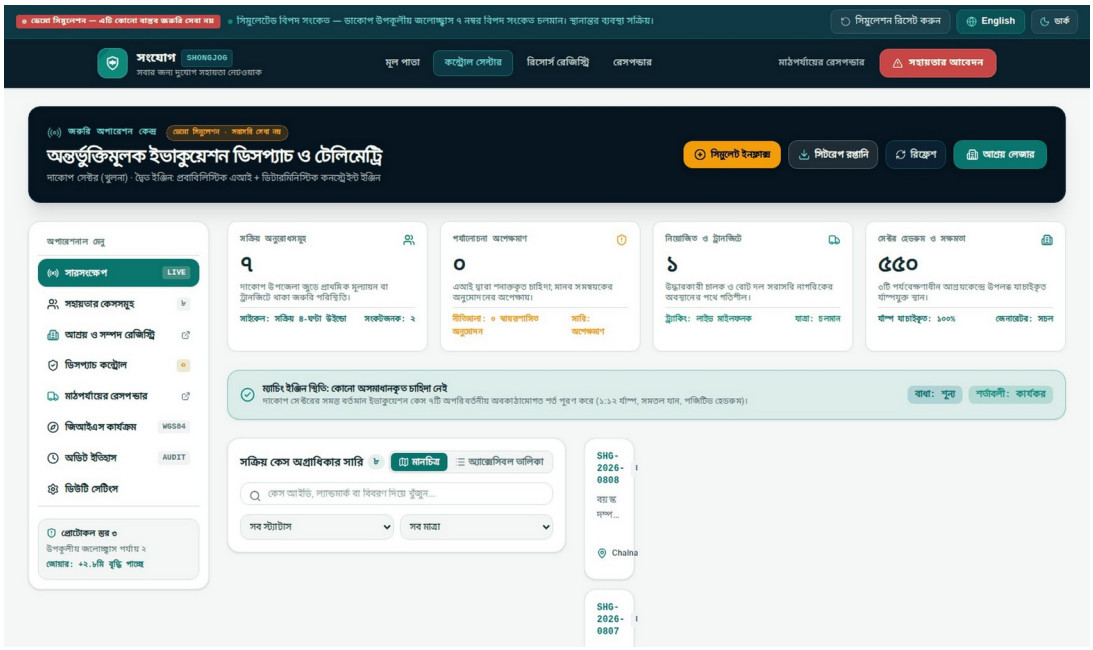
Offline behaviour

The project describes local draft preservation when connectivity is unavailable. A locally saved draft is not a delivered emergency request; the user interface should state this clearly until a successful server sync occurs.

05 / OPERATIONS COMMAND CENTER

A shared workspace for case review and dispatch

The coordinator view brings the assistance queue, resource readiness and geographic context into one operations workflow.



Operations Center screenshot excerpt — Bangla interface.

Main operational areas

OVERVIEW Active requests, review queue, available capacity, responder activity and operational notices.	**ASSISTANCE CASES** Search, filter, inspect a case, review group size, urgency, location and support needs.
RESOURCE REGISTRY Review shelters, vehicles, responders, their attributes, capacity and verification status.	**DISPATCH CONTROL** Compare eligible candidates, inspect constraint explanations and authorize an assignment.
FIELD RESPONDERS View assignments and status progression from acceptance to completion.	**GIS OPERATIONS / AUDIT / SETTINGS** Geographic inspection, case activity history and operational configuration surfaces.

Human-in-the-loop authorization

The coordinator is the approval point for consequential assignments. AI extraction and deterministic matching provide structured information and candidate recommendations; they do not replace the coordinator's decision.

> **Operational caution**
>
> KPI tiles and alerts in the demonstration environment may be seeded or simulated. They must be labeled as demo data unless they are connected to verified live operational feeds.

06 / AI & RESOURCE MATCHING

Probabilistic interpretation. Deterministic eligibility.

SHONGJOG separates language understanding from physical eligibility checks. This separation reduces the risk of a plausible-sounding AI output being treated as a safe dispatch decision.

Dual-engine approach

AI-ASSISTED NEEDS EXTRACTION	DETERMINISTIC MATCHING
Interprets a natural-language request and proposes a structured record: language, group size, support needs, location, urgency, missing information and review requirement.	Queries resources and applies explicit rules to availability, verification, physical access features, capacity and responder skills before ranking viable candidates.

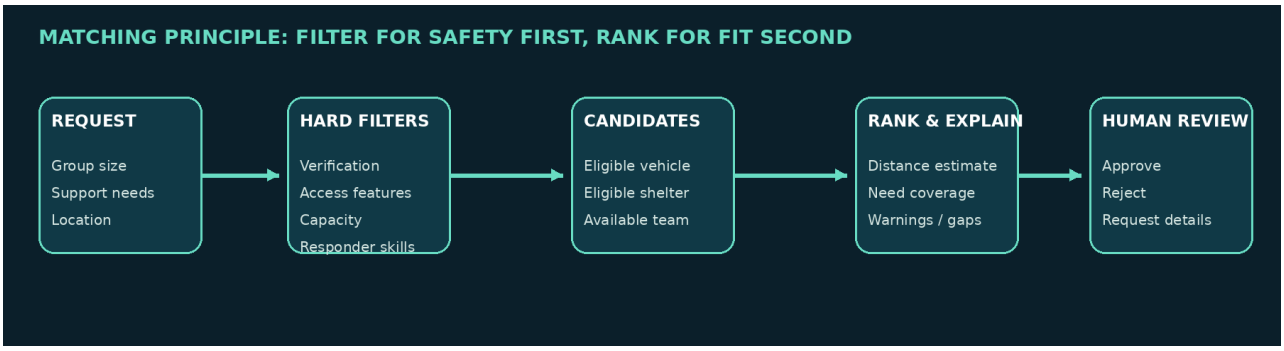


Figure 5. Conceptual matching pipeline: safety/eligibility filters precede ranking and human approval.

Examples of hard constraints in the code

Requirement	Matching behaviour
Wheelchair transport	Reject a vehicle without the required wheelchair-accessible configuration/capacity.
Step-free access	Reject a shelter that does not record step-free access or an available ramp when required.
Power-dependent equipment	Require shelter power-backup capability when the request indicates that need.
Capacity	Reject candidates whose available headroom is below the requested group size.
Verification	Keep resources outside verified / partner-verified tiers out of the eligible matching pool.
Communication / guidance	Prefer available verified responder teams with relevant skills such as sign language or guided assistance.

Distance is not safety

The engine calculates straight-line distance for ranking candidates that already pass its eligibility filters. It does not by itself calculate road, river, weather or route safety. A closer but inaccessible resource should not be selected merely because it is closer.

07 / FIELD RESPONSE & RESOURCE REGISTRY

From assignment to field status

The responder interface focuses on the next operational action and a concise case briefing, while the resource registry stores the capabilities needed by matching.



Accessible boarding / transfer context



Riverine transport context

Responder lifecycle

Stage	Purpose
Assigned	A coordinator has created the assignment and reserved selected resources.
Accepted / declined	The responder confirms availability or declines with a reason.
En route	The team begins the response and can open a Google Maps navigation link.
Arrived	The responder records arrival at the rendezvous or destination.
Completed / unable to assist	The responder closes the task or records why it could not be completed.

Resource registry fields

- Resource identity, type, display name, location and contact/operational notes.
- Verification status and last-verified information.
- Capacity: total, reserved and occupied headroom.
- Capability details: wheelchair access, ramp/step-free access, accessible sanitation, backup power, responder skills.

Audit trail

Case timeline records are designed to capture events, actors, timestamps and operational details. These records support review and traceability; production immutability requires appropriate database permissions and backup controls.

08 / GEOGRAPHIC CONTEXT

Maps help locate work; they do not certify route safety

The geographic workspace is centered on Dacope Upazila, Khulna, using the project reference coordinate 22.5726° N, 89.5124° E.



Coastal delta context used by the project to frame geographic coordination.

Map interaction model

MAP VIEWS Roadmap, Satellite and Terrain presentation modes are included in the map component.	**OPERATIONAL MARKERS** Incidents and resources can be represented by coordinates and selected from the adjacent case/resource list.
SELECTION SYNC Selecting a case or resource updates the map focus and supports inspection of the selected item.	**NAVIGATION HAND-OFF** Responders can open an external Google Maps link for coordinate-based navigation.

Integration modes

When `NEXT_PUBLIC_GOOGLE_MAPS_API_KEY` is configured, the code attempts to load the Google Maps JavaScript API via `@googlemaps/js-api-loader`. The component also constructs an official Google Maps embed URL as a fallback path when a key is absent or the JavaScript loader fails.

Route safety limitation

A coordinate pin or Google Maps navigation deep-link is not proof that a route is passable, accessible, open or safe during a flood/cyclone. Reliable field use requires verified local road/waterway data, current hazard information and human route assessment.

09 / TECHNICAL ARCHITECTURE

A full-stack web application with a persistent data layer

The current GitHub main branch describes a Next.js App Router application with server-side API routes, Prisma data access and PostgreSQL storage for deployment.

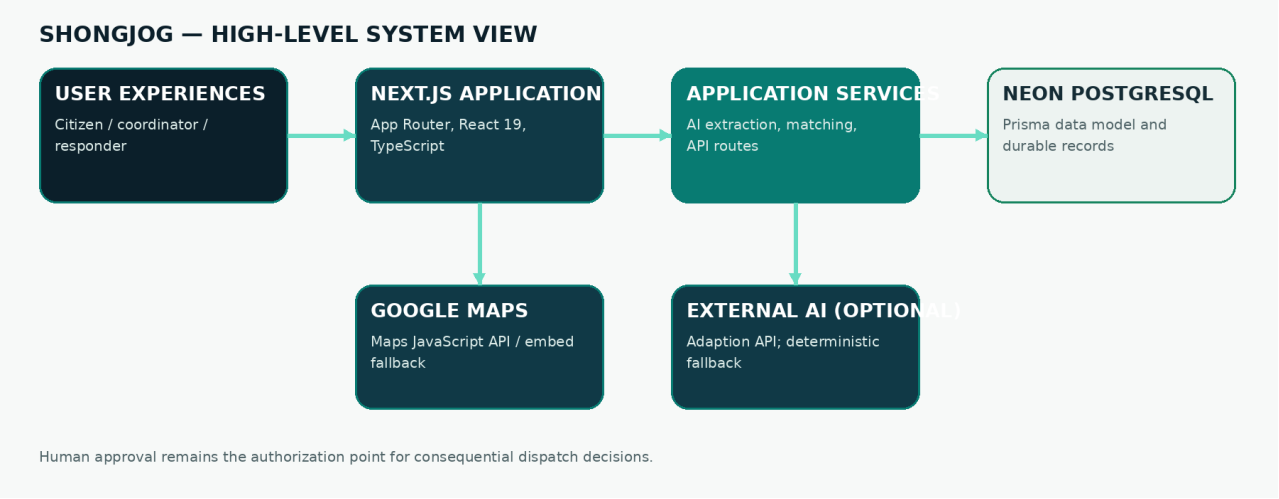


Figure 7. High-level architecture based on current GitHub main source and deployment documentation.

Technology profile

Layer	Technology / role
Web framework	Next.js 16.4.5 App Router; server-rendered and client-interactive routes.
UI runtime	React 19.3.0 with TypeScript.
Styling & motion	Tailwind CSS v4, Lucide icons and Framer Motion.
Validation	Zod schemas at selected API boundaries and typed application models.
Data access	Prisma ORM 6.4.1.
Database	PostgreSQL datasource in current GitHub main; Neon Serverless PostgreSQL documented for production.
Geospatial UI	Google Maps JavaScript API loader plus Google Maps embed/navigation fallback.
Testing	Vitest for unit/integration; Playwright for browser E2E and responsive checks.
Hosting	Vercel production deployment documented at shongjog-eight.vercel.app.

Configuration source of truth

The current GitHub main branch is the production configuration reference. The ZIP source snapshot supplied separately during review contained an older SQLite Prisma schema and `.env.example`; that archive should be regenerated from the production commit before distributing it as the authoritative source package.

10 / DATA & API SURFACE

Structured records for requests, resources and accountability

The Prisma schema on the current GitHub main branch models the core entities needed for intake, matching, assignments and audit history.

Core data entities

Entity	Purpose
AssistanceRequest	Citizen request, urgency, language, group size, description, location and consent.
SupportNeed	Normalized functional needs associated with a request.
Resource	Shelter, vehicle, responder or other resource with status, location and capacity.
VehicleDetail	Vehicle type, wheelchair-accessibility flags, capacity and equipment traits.
ShelterDetail	Step-free access, ramp, accessible toilet and backup-power features.
ResponderDetail	Organization, responder skill profile and operational status.
Assignment	Request-to-resource/responder relationship and status lifecycle.
Reservation	Resource capacity reservation associated with a request.
CaseTimeline	Timestamped actor/event history for a case.
AlertEvent	Scenario/alert record for hazard context.

Application API routes

Route	Purpose
GET /api/health	Health/readiness signal.
GET, POST /api/requests	List and create assistance requests.
GET /api/requests/[id]	Retrieve an individual case and related status/timeline.
GET /api/matching/[requestId]	Return resource matching output for a request.
POST /api/assignments	Create an assignment and reserve capacity transactionally.
POST /api/assignments/[id]/status	Advance or close a responder assignment.
GET, POST /api/resources; PATCH /api/resources/[id]	List/create/update resource records.
POST /api/ai/analyze-needs	Analyze natural-language needs.
POST /api/coordinator/dispatch; POST /api/responder/action	Workflow adapter endpoints.

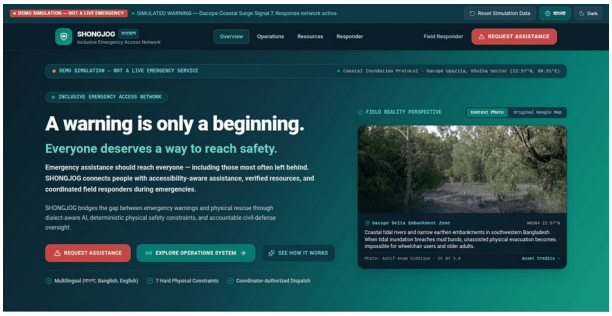
Migration governance

The deployment guide documents a Neon/PostgreSQL setup. For long-lived production data, maintain reviewed, versioned schema migrations and backup/recovery procedures. Do not reset a production database to resolve schema drift.

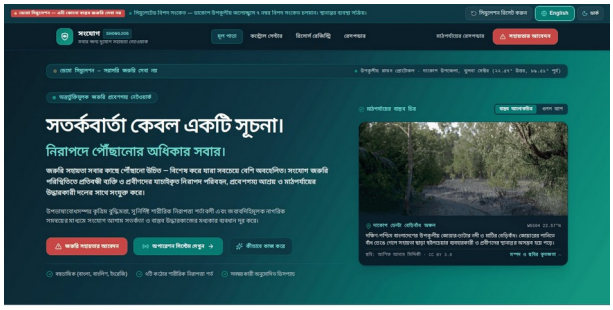
11 / INCLUSIVE UX

English and Bangla across the same product

SHONGJOG is designed as one application with a shared component system and a language preference, rather than separate English and Bangla sites.



English landing experience



Bangla landing experience

Localization implementation

- Central translation dictionary in `src/lib/i18n.ts`, with English (`en`) and Bangla (`bn`) language modes.
- Domain mappers for request status, urgency, functional needs and resource types.
- Pre-paint language initialization and persisted preference via browser storage/cookie, as documented by the release audit.
- Bengali typography stack using Noto Sans Bengali / Hind Siliguri and adjusted line height.
- Locale-aware numeric presentation using `Intl.NumberFormat('bn-BD')`, while preserving ASCII case IDs, URLs and technical notations.

Responsive verification profile

The project release report describes tests at 12 widths from 320 px through 1920 px, including compact phones, tablets, laptops and wide desktop displays. The operations navigation changes to a mobile drawer below the 1280 px breakpoint.

Language / viewport coverage	Reported verification
English	All 12 viewport widths in the responsive suite.
Bangla	The detailed E2E spec checks Bangla overflow at selected boundary widths (320, 390, 768 and 1440 px), plus language switching and route navigation.
UI assertions	Document overflow checks and visibility/bounding-box checks for key controls are described in the release audit.

Coverage precision

The repository's responsive test spec exercises 12 widths in English and checks Bangla at selected widths. The broader release report describes both-language coverage; final acceptance should retain this distinction unless the test matrix is expanded to every width in both languages.

12 / SAFETY, PRIVACY & RESPONSIBLE USE

A coordination tool must be honest about what it knows

Emergency data is sensitive. SHONGJOG's design includes several useful safeguards, but real-world use requires stronger operational controls than a competition demonstration.

Safeguards described in the product

- Functional-needs framing: capture the assistance barrier, not an unnecessary medical diagnosis.
- Consent capture before a request is stored and shared for coordinator review.
- Zod-validated AI result schema with explicit missing-information and human-review fields.
- Deterministic resource filters for capacity, verification and physical feature requirements.
- Human coordinator approval before the assignment is created.
- Transactional resource reservation and assignment state transitions.
- Timeline events for case creation, AI analysis, assignment and responder updates.
- Visible demo/simulation labeling and offline draft messaging.

Production readiness requirements

Risk area	Requirement before real deployment
Identity & authorization	Require authenticated coordinator/responder identities; derive authority server-side, not from an optional client-supplied name or ID.
Personal information	Protect request lists and case details; minimize fields returned to each role and restrict contact/location data.
Resource verification	Define who verifies shelters, vehicles, equipment and responders; record source, date and expiry/reverification.
Database operations	Use durable backups, reviewed migrations, monitoring, restore drills and safe connection pooling.
AI reliability	Log model/fallback mode, preserve source text, validate schemas, show uncertainty and keep human review for ambiguous/high-risk cases.
Field safety	Use verified local route/hazard information and operational protocols; a map pin is not a safe-route guarantee.

Not an official emergency service

SHONGJOG is a software prototype / coordination demonstrator. It does not itself issue official warnings, certify safe shelters, guarantee transport or replace emergency authorities. For emergencies, people must continue to use official emergency channels.

13 / VERIFICATION & DEPLOYMENT

A release profile with evidence, not a guarantee of field readiness

The project includes automated verification assets and a production deployment configuration. The figures below are reported in the repository's V4.2.1 release documentation.

28 / 28 Vitest unit and integration tests reported passing.	**40 / 40** Playwright end-to-end test runs reported passing across four browser/viewport projects.
12 WIDTHS Responsive checks documented from 320 px to 1920 px.	**154 + 154** English and Bangla translation dictionary keys reported with exact parity.

Reported engineering gates

Gate	Reported result	What it covers
TypeScript	0 errors	Static type checking (`npx tsc --noEmit`).
ESLint	0 errors / warnings	Lint checks (`npm run lint`).
Vitest	28 passed	AI/matching, API/integration, concurrency and localization suites.
Playwright	40 runs passed	Five core routes, key journeys and viewport profiles as described in test reports.
Production build	16 routes compiled	Next.js Turbopack build report.
Vercel	Success check on commit 7bc54e7	GitHub commit status linked to Vercel; production URL listed in project documentation.

Deployment profile

Current GitHub main branch uses PostgreSQL in `prisma/schema.prisma` and documents Neon Serverless PostgreSQL with Vercel hosting. The production URL listed by the project is https://shongjog-eight.vercel.app.

The GitHub main branch commit `7bc54e7a77c4f0cc54a71f423892e4da966b5b71` has a successful Vercel status check according to the repository's commit status. The public site was not independently fetched during preparation of this PDF.

Evidence classification

Test counts and deployment details are reported by the project's own release documentation and GitHub status. They are not a substitute for independent penetration testing, live incident drills, field validation or a documented disaster-recovery exercise.

14 / WHAT COMES NEXT

From competition prototype to field-tested coordination

The next stage is not primarily adding more UI. It is validating the workflow with the people, organizations and resource owners who would use it.

1. COMMUNITY CO-DESIGN Work with persons with disabilities, older adults, caregivers and local organizations to test language, intake questions and evacuation workflows.	**2. VERIFIED RESOURCE NETWORK** Partner with shelter managers, transport providers and volunteer groups; define evidence, re-verification and capacity update responsibilities.
3. OPERATIONAL PILOT Run tabletop and controlled field exercises in Dacope with local disaster-management stakeholders before any live use.	**4. CONNECTIVITY RESILIENCE** Evaluate SMS/USSD intake, assisted telephone intake, offline synchronization and retry behaviour in low-connectivity settings.
5. GIS & HAZARD DATA Integrate authoritative, current local infrastructure and hazard layers; distinguish verified routes from simple coordinate navigation.	**6. SECURITY & GOVERNANCE** Complete role-based access, least-privilege data views, threat modelling, retention rules, incident response and audit/backup controls.

Pilot success measures

- Time from request submission to coordinator review.
- Share of cases with all required functional needs confirmed before assignment.
- Resource match acceptance and rejection reasons, including unmatched needs.
- Dispatch completion and response-state update completeness.
- Bangla task completion and accessibility feedback from representative users.
- Data quality: resource verification age, capacity freshness and geolocation accuracy.

> **North-star outcome**
> Measure whether the system helps make appropriate assistance easier to coordinate—not simply the number of requests or dashboard interactions.

15 / REFERENCES & PROJECT LINKS

Evidence, source code and demonstration

This dossier combines repository inspection, the project's release documentation and selected external references. The project-specific implementation claims should be read alongside the cited source files.

External evidence and standards

[1] United Nations Office for Disaster Risk Reduction (UNDRR). 2023 Global Survey Report on Persons with Disabilities and Disasters. Survey overview and results, including the 84% preparedness-plan finding and the global survey scope.

https://www.undrr.org/report/2023-gobal-survey-report-on-persons-with-disabilities-and-disasters

[2] CBM Global Disability Inclusion. Study on disability-inclusive and gender-responsive early warning systems in Khulna, Bangladesh. PreventionWeb publication record, 2026.

https://www.preventionweb.net/publication/documents-and-publications/study-disability-inclusive-and-gender-responsive-early

[3] World Wide Web Consortium (W3C). Web Content Accessibility Guidelines (WCAG) 2.2, Recommendation.

https://www.w3.org/TR/WCAG22/

Project source references

[4] SHONGJOG GitHub repository — current main branch; commit 7bc54e7a77c4f0cc54a71f423892e4da966b5b71. Key files: `src/app/*`, `src/components/GoogleOperationsMap.tsx`, `src/lib/i18n.ts`, `src/lib/matching/engine.ts`, `prisma/schema.prisma`, `docs/V4_2_1_RELEASE_VERIFICATION.md`, `docs/TEST_REPORT.md`.

https://github.com/Safin313-REBORN/SHONGJOG-PLATFORM-OPERATIONAL-DIRECTIVE-FOR-ANTIGRAVITY

Try the project

LIVE DEMO	SOURCE CODE
https://shongjog-eight.vercel.app	github.com/Safin313-REBORN/SHONGJOG-PLATFORM-OPERATIONAL-DIRECTIVE-FOR-ANTIGRAVITY

Prepared as a project dossier

Version: SHONGJOG V4.2.1 | Date: 7 November 2026 | Purpose: competition submission and product overview. This document describes the implementation and intended operating model; it is not a certification of emergency-service readiness.